

COLIN VAUGHN

Meridian, ID 83646



colin-v Vaughn.github.io/portfolio



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SUMMARY

Honors Mechanical and Nuclear Engineering student with experience in R&D bioengineering, robotics, and aerospace systems research. Proficient in CAD, instrumentation integration, and project coordination. Interested in energy system engineering, automation, and advanced mechanical design.

EDUCATION

Honors B.S. MECHANICAL & NUCLEAR ENGINEERING

Minors in Applied Mathematics, Mathematics and Physics

Idaho State University (ISU) College of Science and Engineering
Pocatello, ID | Expected May 2028

Accolades: 3.84 GPA, Dean's List (2024- present), University Honors Program (2024) ISU Presidential Scholarship (2024)

SKILLS & COMPETENCES

Engineering Methods: Agile Development, Bill of Materials Management, Technical Documentation, Design-for-Manufacturing, Microsoft Office Suite

Mechanical Design: SolidWorks, AutoCAD, Mechanical Modeling, Rapid Prototyping

Programming & Computing: HMI Development, MATLAB, C++, Python, HTML/CSS

Controls & Robotics: PID, Feedback-driven Autonomous Systems, Mechatronics

Design & Visualization: Adobe Photoshop, Adobe Illustrator, Adobe Premiere Pro

PROJECT HIGHLIGHTS

- Reel Reactor® R&D phase assembly, programming, and evaluation
- Turbine compression chamber analysis via CAD, prototyping, and coded simulations
- Radionuclide interactive decay chain modeling program in Python
- VEX Robotics PID drive control system & autonomous code

EXPERIENCE

May 2026 – Present **MECHANICAL ENGINEERING INTERN**
Reel Reactor Inc. | Star, ID

- **Mechatronics & Programming:** Programmed a Human-Machine Interface and Python data bridge to enable manual control and runtime autonomous operation of an R&D bioreactor
- **System Optimization:** Participated in rapid prototyping, design-for-manufacturing workflows, and technical reviews with senior engineers to evaluate and improve designs, maintain a bill of materials
- **Business Operations:** Presented technical milestones and progress updates in investor meetings to support ongoing project fundraising

Feb 2026 – Present **AEROSPACE SYSTEMS RESEARCH LAB MEMBER**
Measurement & Control Engineering Research Center | Pocatello, ID

- **Machine Design:** Designing and fabricating testing interfaces to simulate pressure accumulation in a jet turbine and evaluate optimized compression chamber geometry
- **Instrumentation Integration:** Incorporated systems for iterative data collection and performance evaluation
- **Project Coordination:** Synchronized efforts in a multidisciplinary team of undergraduates, graduates and faculty

Oct 2021 – Apr 2024 **COMPETITIVE ROBOTICS TEAM CAPTAIN**
VEX Robotics Competition Team | Meridian, ID

- **Agile Project Development:** Directed iterative subsystem refinement involving drivetrains, sensors, manipulators, and PID control programming for regional and international team competitions
- **Technical Documentation:** Maintained a multi-volume, award-winning engineering notebook documenting CAD models, design iterations, and project budgeting
- **Team Leadership:** Managed an 8-member engineering team through full development cycles, coordinating task allocation and cross-functional collaboration

ADDITIONAL EXPERIENCE

SHIFT SUPERVISOR | *Jersey Mikes Subs | Meridian, ID | May 2024 – Aug 2025*

LEAD GAME DESIGNER & BUILDER | *Self-Started "Rivals Gaming" | Meridian, ID | Jun 2021 – Dec 2023*